

ADP470S Practice Assessment: Data Structures & Algorithms

Instructions: Focus on illustrating your thought process. Use diagrams, memory state snapshots, and tree visualizations to support your answers.

Question 1: Visualizing Recursion [15 Marks]

1.1 Given a recursive factorial function, draw the complete call stack diagram (showing each push and pop) for the execution of `factorial(4)`. Clearly label the return values cascading back up the stack at each step. [10]

1.2 Briefly explain why this specific recursive stack trace demonstrates $O(n)$ space complexity. [5]

Question 2: Asymptotic Analysis [15 Marks]

Determine the Time Complexity (Big-O) of the following snippets. Show your working steps.

2.1

```
int count = 0;
for (int i = 1; i < n; i = i * 2) {
    for (int j = 0; j < n; j++) {
        count++;
    }
}
```

[8]

2.2

```
for (int i = 0; i < n; i++) {
    // some O(1) operation
}
for (int j = 0; j < n; j++) {
    for (int k = 0; k < n; k++) {
        // some O(1) operation
    }
}
```

[7]

Question 3: Sorting Traces [25 Marks]

3.1 HeapSort: Draw the initial Max-Heap tree structure for the following array: {15, 3, 18, 10, 22, 5}. Then, draw the state of the tree after the first maximum element is extracted and the remaining elements are heapified. [12]

3.2 QuickSort: To demonstrate the worst-case scenario, trace the partitioning steps of QuickSort on an already sorted array: {10, 20, 30, 40, 50}, assuming the last element is always chosen as the pivot. Use diagrams to show the sub-arrays at each step. [13]

Question 4: Circular Queue States [20 Marks]

Assume a Circular Queue of capacity 4 (indices 0 to 3), initially empty, where both Head and Tail start at index 0.

Draw the array and the positions of the Head and Tail pointers after the following sequence of operations:

1. Enqueue(A)
2. Enqueue(B)
3. Enqueue(C)
4. Dequeue()
5. Enqueue(D)
6. Enqueue(E)
7. Dequeue()

Provide a snapshot diagram of the queue state immediately after step 3, step 5, and step 7. [20]

Question 5: Binary Search Tree Mapping [25 Marks]

5.1 Draw the final Binary Search Tree after inserting the following elements into an empty tree in this exact order: {50, 30, 70, 20, 40, 60, 80}. [10]

5.2 For the tree you just drew, write out the exact sequence of nodes visited during a Pre-order Traversal and a Post-order Traversal. [10]

5.3 Draw the new state of the tree after inserting the value 35. [5]